

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and optimized **large-scale AI model pre-training pipelines** across distributed **GPU infrastructures** using **PyTorch** and **TensorFlow**, achieving **40% improvement** in training throughput and reducing computational bottlenecks across multi-node clusters at HCL Technologies.
- Researched and prototyped **advanced transformer architectures** and **large language model** optimizations, conducting independent and collaborative experiments to analyze training results, iterate on methodologies, and improve model quality and efficiency by **35%** on key downstream benchmarks.
- Enhanced **distributed training systems** for seamless scalability, reliability, and operational efficiency across large compute environments, implementing **parallel training strategies** and **infrastructure optimization** techniques that reduced training time by **28%** while maintaining model convergence stability.
- Identified, investigated, and resolved **performance bottlenecks** in model scaling and computational optimization using **deep learning optimization** methodologies and **model scaling** best practices, conducting rigorous analytical and problem-solving work to enhance GPU utilization by **32%**.
- Contributed to **data curation strategies** and baseline improvements using **Python** scripting, data pipeline automation, and **distributed computing** techniques to strengthen model training outcomes, implementing ETL workflows that processed **500GB+ daily** training datasets with **99.8% accuracy**.
- Collaborated cross-functionally with engineering and research teams across remote and international environments to accelerate AI research innovation, staying informed on emerging trends in machine learning systems and large-scale model training while maintaining **research-driven** experimentation practices and strong communication within globally distributed teams.

SKILLS

Large Language Models & AI Research – transformer architectures, large language model pre-training, multi-modal AI systems, advanced architecture design, model intelligence optimization, efficiency and scalability research.

Machine Learning Frameworks – PyTorch, TensorFlow, JAX, scikit-learn, deep learning optimization, model scaling, performance tuning.

Distributed Computing & GPU Training – distributed machine learning systems, large-scale GPU-based training environments, parallel training strategies, distributed computing, infrastructure optimization, multi-node cluster management.

Python & Systems Development – Python, scalable AI training systems, data pipeline development, ETL automation, shell scripting, Linux systems administration.

Data Engineering & Curation – data curation strategies, baseline improvements, data modelling, data warehousing, SQL (PostgreSQL, MySQL, Oracle), distributed data processing.

Cloud & DevOps Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD pipelines, Git, Jenkins, change-control management.

Research & Experimentation – research-oriented mindset, experimental design, analytical problem-solving, performance analysis, computational bottleneck resolution, innovation and novel technique exploration.

EDUCATION AND TRAINING

PROJECTS

Large-Scale Transformer Pre-training Pipeline Optimization (HCL Technologies)

- Architected and implemented a **distributed GPU pre-training pipeline** for a **multi-modal language model** spanning **1,024 A100 GPUs** across 8 nodes, utilizing **PyTorch Distributed Data Parallel** and **FSDP (Fully Sharded Data Parallel)** to enable training of **70B parameter models** with **43% reduction** in training time.
- Optimized training methodologies by researching and implementing advanced **gradient accumulation**, **mixed-precision training**, and **activation checkpointing** strategies, improving **GPU memory efficiency** by **38%** and enabling batch sizes **2.5x larger** without out-of-memory failures.
- Developed robust **data curation and preprocessing pipelines** using **Python**, **pandas**, and **Apache Spark** to process **2.3TB of raw training data** daily, implementing deduplication, quality filtering, and tokenization that improved downstream model convergence by **22%** and reduced training instability.
- Designed comprehensive monitoring and observability for **distributed training systems** using **AWS CloudWatch** and **Weights & Biases**, capturing real-time metrics on GPU utilization, loss trajectories, and communication latencies across clusters, enabling rapid identification and resolution of **13 critical bottlenecks**.
- Collaborated with cross-functional research teams to prototype **novel transformer architecture variants** and conduct ablation studies, executing **40+ independent experiments** to validate scaling laws and model efficiency improvements, with findings contributing to **18% FLOPS reduction** without accuracy degradation.

Payment Systems Data Pipeline & Distributed Training Infrastructure (HCL Technologies – Banking Domain)

- Engineered high-throughput **distributed data pipelines** for payment-operations training datasets using **Python**, **SQL** (PostgreSQL, Oracle), and **AWS Lambda**, processing **SWIFT MT/MX message flows** and **ISO 20022** payment structures to curate **500M+ transaction records daily** with **99.95% data integrity**.
- Optimized **infrastructure scaling** across production GPU environments using **Docker containerization**, **Kubernetes orchestration**, and **distributed computing** principles, achieving **auto-scaling efficiency of 41%** and reducing infrastructure costs by **\$180K quarterly** while maintaining **99.99% SLA uptime**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)